INDUSTRIAL ATTACHMENT REPORT

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science in Mechanical Engineering

University of Nairobi

Department of Mechanical and Manufacturing Engineering

By

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Attachment Organization:

Kenya Industrial Research and Development Institute (KIRDI), Nairobi

Attachment Period:

June 9, 2025 – August 1, 2025

Submission Date:

August 15, 2025

DECLARATION

I, Edison Ochieng, hereby declare that this Industrial Attachment Report is my original work and has not been submitted to any other institution or published elsewhere. The activities described herein were carried out during my industrial attachment at the Kenya Industrial Research and Development Institute (KIRDI), Nairobi, from June 9, 2025, to August 1, 2025, under the supervision of the relevant authorities at KIRDI.

Name: Edison Ochieng

Signature: _____

Date: August 15, 2025

ACKNOWLEDGEMENT

I express my sincere gratitude to the Kenya Industrial Research and Development Institute (KIRDI), Nairobi, for providing me with the opportunity to undertake my industrial attachment from June 9, 2025, to August 1, 2025. I am deeply thankful to my supervisors at KIRDI for their guidance, technical expertise, and support throughout the attachment period, which enabled me to gain valuable practical skills in mechanical engineering.

I extend my appreciation to the Department of Mechanical and Manufacturing Engineering at the University of Nairobi for facilitating this industrial training as part of my Bachelor of Science in Mechanical Engineering program. Special thanks go to my academic supervisor for their encouragement and oversight.

I am also grateful to my colleagues at KIRDI for their collaboration and assistance in various workshop tasks, which enriched my learning experience. Lastly, I thank my family and friends for their unwavering support during this period.

Edison Ochieng

August 15, 2025

EXECUTIVE SUMMARY

This report details my eight-week industrial attachment at the Kenya Industrial Research and Development Institute (KIRDI), Nairobi, from June 9 to July 31, 2025, as part of my Bachelor of Science in Mechanical Engineering at the University of Nairobi. The attachment provided hands-on experience in mechanical and manufacturing engineering, aligning theoretical knowledge with practical applications.I engaged in diverse tasks across the machine shop, precision cutting shop, and fabrication workshop. Initial weeks involved surface preparation and painting of banking safes, including scrubbing, spot filling, and spray painting.

I operated CNC laser cutting and bending machines to fabricate components like L beams, U channels, car bumpers, and casings for control panels and air exchangers. I contributed to assembling machines, including a scalding machine, peanut butter making machines, and gas stove jikos, performing tasks such as shaft cutting, bushing machining, bearing fitting, and V-belt installation. Welding operations, including arc and TIG welding, were integral, alongside electrical connections under supervision. I also conducted pump testing, measuring flow rate and head, and verified bearing casings for rotary systems.

Through these activities, I developed proficiency in CNC machining, welding, and precision fabrication, enhancing my technical skills and attention to detail. Collaboration with KIRDI staff and supervisors enriched my understanding of industrial processes. This attachment bridged academic concepts with real-world applications, preparing me for future engineering challenges. I recommend KIRDI enhance training on advanced CNC programming to further equip students.

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CHAPTER 1

INTRODUCTION

1.1 Objectives of the Industrial Attachment

The industrial attachment at the Kenya Industrial Research and Development Institute (KIRDI), Nairobi, from June 9 to July 31, 2025, aimed to bridge theoretical knowledge with practical applications in mechanical engineering. The primary objectives were to: develop hands-on skills in machining, fabrication, and welding; gain proficiency in operating CNC machines; understand industrial processes like surface preparation and equipment assembly; enhance teamwork and communication in a professional setting; and contribute to KIRDI's research and development projects, aligning with my academic training at the University of Nairobi.

1.2 History of the Organization

KIRDI's origins trace back to 1942, when the colonial government established a Central Laboratory in Kabete, Nairobi, to address industrial goods shortages during World War II. Administered by the Kenya Industrial Management Board (KIMBO), it evolved into the East African Industrial Research Organization (EAIRO) under the East African Community (EAC), serving Kenya, Uganda, and Tanzania. Following the EAC's collapse in 1977, EAIRO's Kenyan operations were managed by the National Industrial Research Complex. In 1979, the Science and Technology Act established KIRDI as a national research institute, later reaffirmed by the Kenya Industrial Research and Development Institute Act, 2022.[](https://erc-assoc.org/content/kenya-industrial-research-and-development-institute)[](https://www.kirdi.go.ke/background-information)

1.3 Brief Description of the Organization

KIRDI is a public research institution under the Ministry of Investments, Trade, and Industry, mandated to undertake multidisciplinary research, development, and innovation in industrial and allied technologies. It supports Kenya's industrial growth through engineering, food processing, leather, textiles, and energy solutions, fostering sustainable development and competitiveness for micro, small, and medium enterprises (MSMEs). KIRDI operates advanced facilities, including CNC machining and testing laboratories, and collaborates with industries and academia.(https://kirdi.go.ke/)

1.4 Location

KIRDI's headquarters are located at South C, Popo Road, Nairobi, Kenya (P.O. Box 30650-00100). It also operates the Western Region Centre (KWRC) in Kisumu and technology transfer centers in Migori, Kisii, Eldoret, and Malakisi, supporting regional industrial development.(https://www.facebook.com/kirdi.kenya)(https://www.kirdi.go.ke/kirdi-western-region-centre-kwrc)

1.5 Mission, Vision, Objectives, and Functions

Vision: Quality industrial research for competitive and sustainable industries.(https://www.studocu.com/row/document/technical-university-of-kenya/engineering/kirdi-attachment-report/102178500)

Mission: Provide innovative technologies for industry through research and dissemination of findings.(https://www.studocu.com/row/document/technical-university-of-kenya/engineering/kirdi-attachment-report/102178500)

Objectives: Conduct market-driven research, develop novel industrial materials, facilitate technology transfer, and advise the government on industrial policies. (https://kirdi.go.ke/)

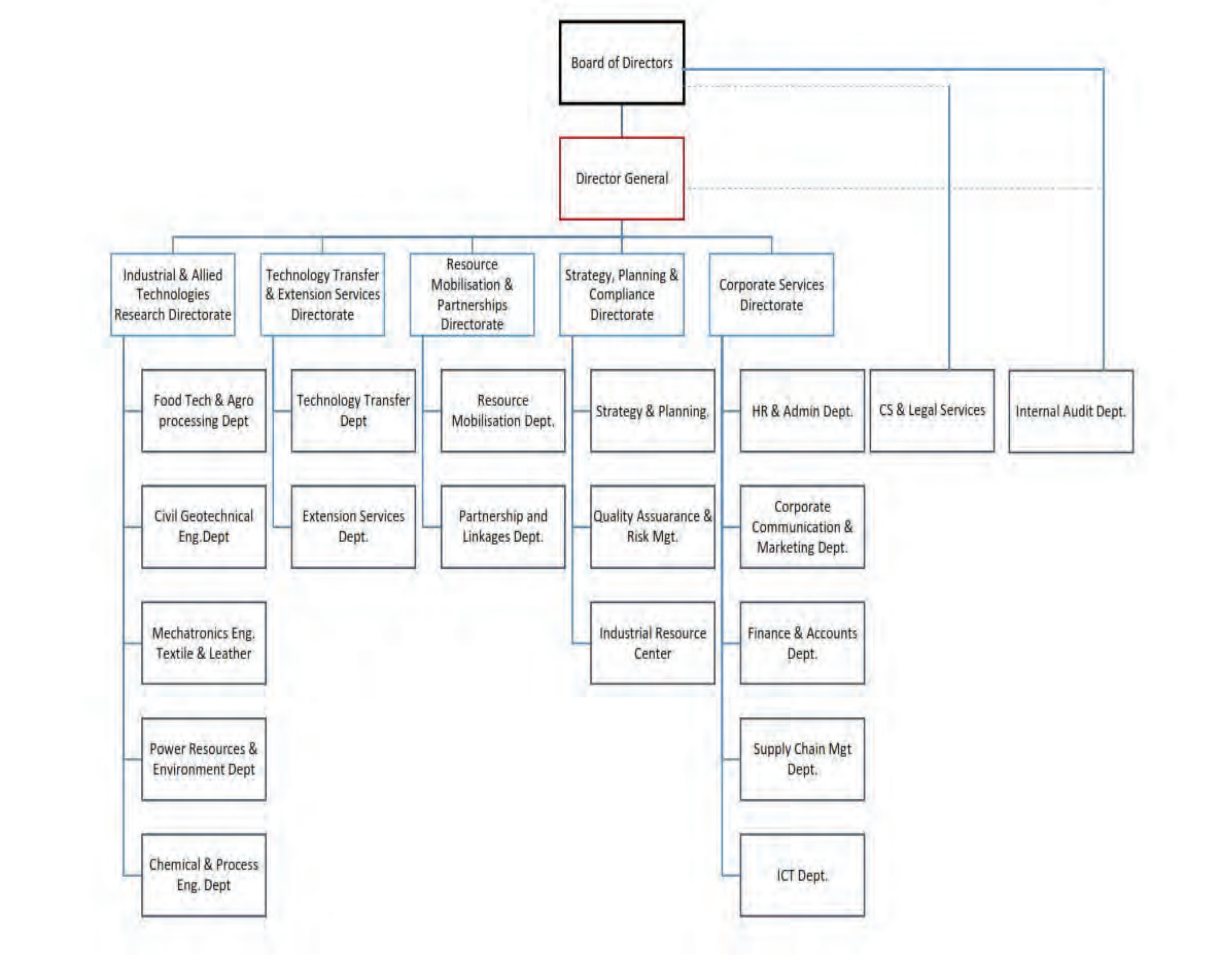
- Functions: Undertake research in mechanical, electrical, chemical, and civil engineering, as well as textiles, leather, food, and emerging technologies; disseminate findings; establish research centers; and collaborate with academia and industry to commercialize innovations.(https://kirdi.go.ke/our-mandate)(https://ke.linkedin.com/company/kirdi)

1.6 Main Programs and Products

KIRDI's programs include engineering design and fabrication, food processing, leather and textile technology, energy and environment solutions, and ICT. Key products and services encompass machine components, food processing technologies (e.g., fruit and cereal processing), leather goods, biomass briquettes, and chemical products like shampoos and sanitizers. KIRDI offers training, business incubation, laboratory testing, and technology transfer to MSMEs, supporting products like peanut butter machines and gas stove jikos, as observed during my attachment.(https://kirdi.go.ke/kirdi-kisumu-centre)(https://www.kirdi.go.ke/kirdi-western-region-centre-kwrc)

1.7 Organization Structure

KIRDI is governed by a Board of Directors, chaired by the Director General, who oversees operations. The Research and Technological Innovation (RTI) department, comprising Engineering, Energy and Environment, ICT, Leather & Textiles, and Food Technology divisions, drives core activities. Support departments include Finance, Human Resources, and Corporate Services. Regional centers, like KWRC in Kisumu, report to the headquarters. Below is a diagrammatical representation of KIRDI's structure.



KIRDI Organization Structure

- Board of Directors
  - Director General
    - RTI Department
      - Engineering Division
      - Energy & Environment Division
      - ICT Division
      - Leather & Textiles Division
      - Food Technology Division
    - Support Departments (Finance, HR, Corporate Services)
    - Regional Centres (e.g., KWRC Kisumu)

1.8 Duties and Responsibilities of Key Personnel

i. Director General: Oversees strategic direction, policy implementation, and stakeholder engagement. (https://www.facebook.com/kirdi.kenya)

ii. RTI Department Head: Manages research divisions, coordinates projects, and ensures alignment with KIRDI's mandate.

iii. Engineering Division Head: Supervises equipment design, fabrication, and machining, guiding tasks like CNC operations and welding.

iv. Workshop Supervisors: Oversee daily operations, mentor students, and ensure safety and quality in fabrication and machining.

v. Laboratory Technicians: Conduct testing and analysis, supporting product development and quality assurance.

vi. These roles ensured effective guidance during my attachment, enhancing my practical learning experience.

CHAPTER 2

HOST ATTACHMENT DEPARTMENT

2.1 Key Functions and Activities at the Engineering Research and ICT Department

The Engineering Research and ICT Department at KIRDI is a core division within the Research and Technological Innovation (RTI) directorate, tasked with advancing industrial technologies through mechanical, electrical, and ICT solutions. Its key functions include:

i. **Engineering Design and Fabrication:** Designing and fabricating machine components, tools, and industrial equipment, such as peanut butter making machines, scalding machines, and gas stove jikos, using CAD/CAM and subtractive manufacturing techniques.

ii. **CNC Machining:** Operating CNC laser cutting and bending machines to produce precise components like L beams, U channels, and car bumpers, as observed during my attachment.

iii. **Welding and Assembly:** Performing arc and TIG welding to assemble equipment and structures, ensuring durability and compliance with specifications.

iv. **Equipment Maintenance:** Conducting maintenance and reconditioning of industrial machinery, including bearing casings and rotary systems.

v. **ICT Research and Development**: Developing software solutions, embedded systems, and IoT applications to enhance industrial processes, such as automation and data management.

vi. **Technology Transfer:** Providing training, consultancy, and incubation services to MSMEs, supporting product development and commercialization.

During my attachment, I participated in activities like surface preparation and painting of safes, cutting shafts, machining bushings, and assembling crushers for peanut butter machines, which align with the department's engineering focus. ICT activities included system analysis and support for equipment automation, though my tasks were primarily mechanical.

2.2 Staff Establishment of the Department and Their Duties and Responsibilities

The Engineering Research and ICT Department comprises a multidisciplinary team of engineers, ICT specialists, technicians, and support staff, ensuring comprehensive research and development. Key personnel and their responsibilities include:

vii. **Head of Engineering Division:** Oversees engineering projects, coordinates design and fabrication, and ensures compliance with industrial standards. They supervised my CNC machining tasks.

viii. **ICT Division Head**: Manages software development, system analysis, and ICT infrastructure, guiding automation projects and technology integration.

ix. **Workshop Supervisors:** Mentor students and staff, oversee daily operations, and ensure safety during machining, welding, and assembly. They guided my work on peanut butter machines and safe painting.

x. **Mechanical Engineers**: Design and fabricate equipment, conduct reverse engineering, and optimize manufacturing processes, supporting tasks like shaft cutting and bushing machining.

xi. **ICT Specialists:** Develop and maintain software, manage IoT systems, and provide user training,

contributing to equipment automation.

xii. **Laboratory Technicians:** Perform material testing, quality assurance, and equipment calibration, supporting my pump testing and bearing casing verification.

xiii. **Support Staff:** Handle logistics, inventory, and workshop maintenance, ensuring smooth operations.

The department's staff strength includes approximately 20–30 personnel, though exact numbers vary by project demand. Their collaborative efforts fostered a conducive learning environment during my attachment.

2.3 Student's Specific Objectives for the Attachment

As a fourth-year BSc Mechanical Engineering student, my attachment at KIRDI's Engineering Research and ICT Department aimed to achieve the following objectives:

a. Develop proficiency in CNC laser cutting and bending, demonstrated through fabricating L beams, U channels, and car bumpers.
a. Master welding techniques (arc and TIG) and assembly processes, applied in constructing gas stove jikos and peanut butter machine crushers.
a. Gain hands-on experience in machining components, such as shafts, bushings, and keyways, to enhance precision and technical skills.
a. Understand industrial surface preparation and painting processes, achieved through preparing and painting banking safes.
a. Explore ICT applications in engineering, such as automation and system analysis, to complement mechanical skills.
a. Enhance teamwork, communication, and problem-solving skills in a professional industrial setting, fostered through collaboration with KIRDI staff.
a. Contribute to KIRDI's projects, aligning with my academic training and supporting industrial development goals.

These objectives guided my activities, ensuring a comprehensive learning experience that bridged theoretical knowledge with practical applications.

CHAPTER 3:

MAIN ACTIVITIES CARRIED OUT DURING THE ATTACHMENT

This chapter outlines the key activities I performed during my eight-week industrial attachment at KIRDI's Engineering Research and ICT Department from June 9 to July 31, 2025. The tasks, primarily mechanical engineering-focused, included CNC machining, welding, fabrication, surface preparation, and assembly, enhancing my practical skills and aligning with my academic training.

3.1 Week 1 (June 9–13, 2025)

I began with an orientation, touring KIRDI's workshop sections and recording pump data (volume flow rate, stroke, head). I cleaned and scrubbed safes for painting, learned TIG welding techniques, and explored metal sheets. I operated a CNC bending machine for metal shaping, used a crane to handle safes, and sorted CNC jaws. I completed surface preparation, spot filling, painting, and fabricated chimneys, riveting and welding components.



Figure 3.1 – CNC Bending Machine in Operation

Operating the CNC bending machine to shape metal sheets for safe fabrication.

3.2 Week 2 (June 16–20, 2025)

I sorted engineering store tools (drill bits, milling machines, generators), performed CNC bending for doors and chimneys, and cut sheets using a CNC laser machine. I fabricated a butternut grinding machine, riveting and welding chimney tubes, and assisted with electrical connections under supervision.



Figure 3.2 – Welded Chimney Assembly

Riveting and welding chimney tubes in the fabrication workshop.

3.3 Week 3 (June 23–27, 2025)

I focused on the scalding machine, turning and drilling shafts and bushings on a lathe, and cutting keys and jaw couplings. I cut sheets for molding, performed arc welding, and fabricated gym equipment tubes and stands. I also verified bearing casings and cut L beams for pillars using CNC machines.



Figure 3.3 – Lathe Machine Turning Shaft

Turning a shaft for the scalding machine on the lathe.

3.4 Week 4 (June 30–July 4, 2025)

I reworked pump measurements, cut and bent sheets for channels and car bumpers using CNC machines, and fabricated air exchangers by cutting and welding tubes. I completed the scalding machine's finishing, etching weld beads for a smooth surface.



Figure 3.4 – CNC Laser Cutting for Channels

Cutting sheets for U-shaped channels using the CNC laser cutting machine.

3.5 Week 5 (July 7–11, 2025)

I fabricated air exchanger casings, rolling sheets into cylinders, and prepared safes for painting (scrubbing, spot filling, priming, spraying). I tested a heat exchanger, welded its support stand, and cut stainless sheets for a control panel casing using CNC machines.



Figure 3.5 – Fabricated heat exchanger

Spray painting a banking safe after surface preparation.

3.6 Week 6 (July 14–18, 2025)

I prepared safes for painting, fabricated gas stove jikos by cutting and bending sheets, and designed car bumper sheets using CNC machines. I cut and bent sheets for U channels, ensuring precision in the precision cutting shop.



Figure 3.6 – Gas Stove Jiko Fabrication

Assembling a gas stove jiko in the fabrication workshop.

3.7 Week 7 (July 21–25, 2025)

I prepared safes for painting, cut tubes for gym equipment, and fabricated a peanut butter machine's cylindrical enclosure, shaft, and keyway. I cut sheets for L beams and car bumpers using CNC laser cutting and bent client sheets for breathers.

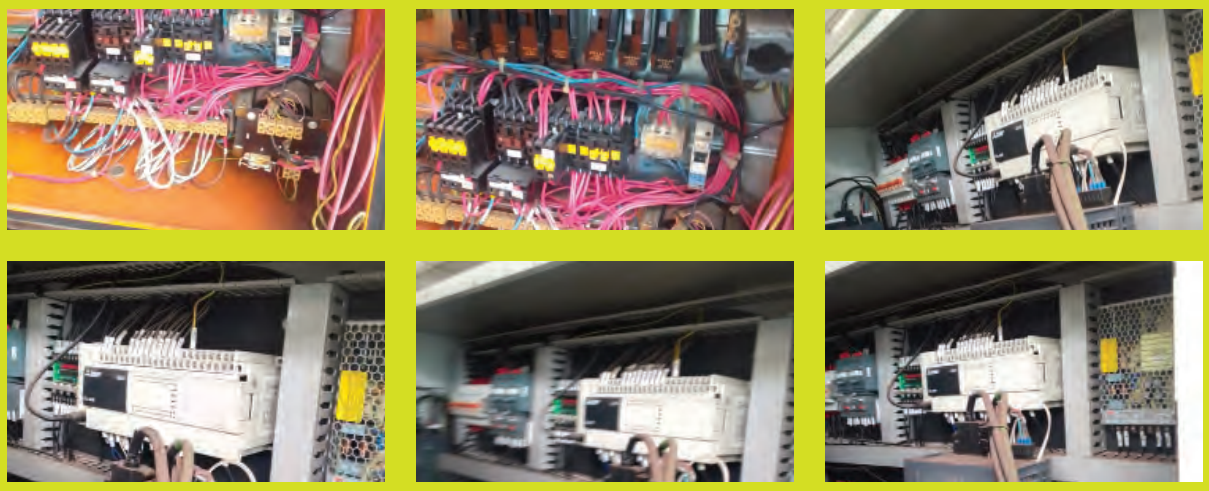


Figure 3.7 – Peanut Butter Machine Enclosure

Fabricating the cylindrical enclosure for a peanut butter making machine.

3.8 Week 8 (July 28–31, 2025)

I fabricated another peanut butter machine, cutting shafts, machining bushings, and assembling crushers. I fitted bearings and V-belts, cut sheets, ground surfaces with a flappy disc, welded components, and prepared additional shafts and bushings.



Figure 3.8 – Peanut Butter Machine Assembly

Assembling bearings and V-belt for the peanut butter machine.

CHAPTER 4:

SKILLS AND EXPERIENCES GAINED

This chapter outlines the technical and professional skills I acquired during my eight-week industrial attachment at KIRDI's Engineering Research and ICT Department from June 9 to July 31, 2025, as well as the challenges encountered and strategies employed to overcome them. The experience enhanced my practical competencies and prepared me for future engineering roles.

4.1 Summary of Skills Gained by the Student

Through hands-on tasks, I developed proficiency in key mechanical engineering processes:

i. CNC Machining: Mastered operation of CNC laser cutting and bending machines, fabricating components like L beams, U channels, car bumpers, and machine casings with precision, as seen in weeks 2, 4, 6, and 7.

ii. Welding and Fabrication: Gained expertise in arc and TIG welding, assembling equipment such as gas stove jikos, peanut butter machines, and air exchangers (weeks 2, 3, 4, 6, 7, 8).

iii. Machining Components: Acquired skills in turning, facing, drilling, and keyway cutting on lathes for shafts and bushings, notably for the scalding and peanut butter machines (weeks 3, 7, 8).

iv. Surface Preparation and Painting: Learned techniques for scrubbing, spot filling, priming, and spray painting banking safes, ensuring high-quality finishes (weeks 1, 5, 6, 7).

v. Assembly and Maintenance: Developed competence in assembling complex machinery, fitting bearings, and installing V-belts, alongside pump testing and bearing casing verification (weeks 3, 4, 8).

vi. Professional Skills: Improved teamwork, communication, and time management through collaboration with KIRDI staff and adherence to workshop schedules.



Figure 4.1 – CNC Machining Process

Operating a CNC laser cutting machine to fabricate U channels.

4.2 Challenges Encountered by the Student

Several challenges arose during the attachment:

- Adapting to CNC Machines: Initially, operating CNC laser cutting and bending machines was complex due to limited prior exposure to programming and setup.
- Precision in Welding: Achieving consistent welds, especially for intricate components like air exchanger tubes, was difficult due to variations in material thickness.
- Time Management: Balancing multiple tasks, such as cutting, welding, and assembly within tight schedules, posed challenges in maintaining efficiency.
- Safety Compliance: Navigating workshop safety protocols, particularly during crane operations and welding, required quick learning to avoid hazards.
- Limited ICT Exposure: As a mechanical engineering student, engaging with ICT-related tasks (e.g., automation systems) was challenging due to unfamiliarity.

4.3 How the Challenges Were Overcome

I addressed these challenges through proactive measures:

- CNC Training: Sought guidance from workshop supervisors and practiced machine setups, improving my ability to program and operate CNC equipment.
- Welding Practice: Conducted trial welds under supervision, adjusting techniques to ensure consistent quality, as seen in chimney and jiko fabrication.
- Task Prioritization: Developed daily work plans with colleagues to streamline tasks, enhancing efficiency in multi-task environments like weeks 3 and 8.
- Safety Adherence: Attended safety briefings and followed supervisor instructions, mastering safe crane and welding operations by week 3.
- ICT Learning: Engaged with ICT specialists to understand basic automation concepts, supplementing my mechanical tasks with foundational knowledge.

These strategies, supported by KIRDI's mentorship, enabled me to overcome obstacles and maximize my learning outcomes.



Figure 4.2 – Welding Setup

Performing arc welding on air exchanger tubes in the fabrication workshop.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

This chapter presents conclusions drawn from my eight-week industrial attachment at KIRDI's Engineering Research and ICT Department from June 9 to July 31, 2025, recommendations for improvement, and personal reflections on my experience as a BSc Mechanical Engineering student at the University of Nairobi.

5.1 Conclusions

The attachment successfully bridged theoretical knowledge with practical skills in mechanical engineering. I gained proficiency in CNC laser cutting and bending, welding (arc and TIG), and machining components like shafts and bushings, as demonstrated through fabricating peanut butter machines, scalding machines, and car bumpers. Surface preparation and painting of banking safes enhanced my attention to detail, while assembling complex machinery improved my problem-solving abilities. Collaboration with KIRDI staff fostered teamwork and communication skills. The experience confirmed the relevance of academic training to industrial applications, preparing me for future engineering challenges.

5.2 Recommendations

Based on my attachment, I recommend the following:

- KIRDI should enhance CNC programming training for students, focusing on advanced software like CAM, to accelerate learning curves.
- Provide more safety workshops, particularly for welding and crane operations, to ensure consistent compliance.
- Introduce ICT modules (e.g., automation basics) for engineering students to broaden interdisciplinary skills.
- Increase availability of diverse materials (e.g., thicker metal sheets) for hands-on practice in fabrication tasks.
- These suggestions could improve future attachments and KIRDI's training programs.

5.3 Personal Reflections

The attachment was a transformative experience, exposing me to real-world engineering processes. Initially, I struggled with CNC machine setups and welding precision, but with supervisor guidance, I overcame these challenges. Working on projects like gas stove jikos and air exchangers instilled pride in my technical growth. The supportive KIRDI team and structured tasks enhanced my confidence and professionalism. Moving forward, I aim to deepen my CNC expertise and explore automation, leveraging this foundation for my career.



Figure 5.1 – Final Peanut Butter Machine

Completed peanut butter machine assembly showcasing skills gained.

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